

Cryptocurrency Deep Dive

Digital Money and Its Evolution

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Algorithm Basics and BW

What We'll Cover Today

1.  What is cryptocurrency?
2.  Evolution from physical money to crypto
3.  Key features and characteristics
4.  Benefits and drawbacks
5.  Hands-on: Comparing payment systems

Quick Recap: Blockchain Basics

Last class we learned:

- ▶  Blockchain = Shared digital ledger
- ▶  Immutable (can't change past records)
- ▶  Decentralized (no single controller)
- ▶  Smart contracts automate agreements

Today: How cryptocurrency uses blockchain technology

What is Cryptocurrency?

Simple Definition:

- ▶  Digital money that exists only online
- ▶  Secured by cryptography (complex math)
- ▶  Can be sent directly person-to-person
- ▶  No bank or government required
- ▶  Works across borders instantly

Think of it like:

- ▶ Email for money
- ▶ Cash, but digital and borderless

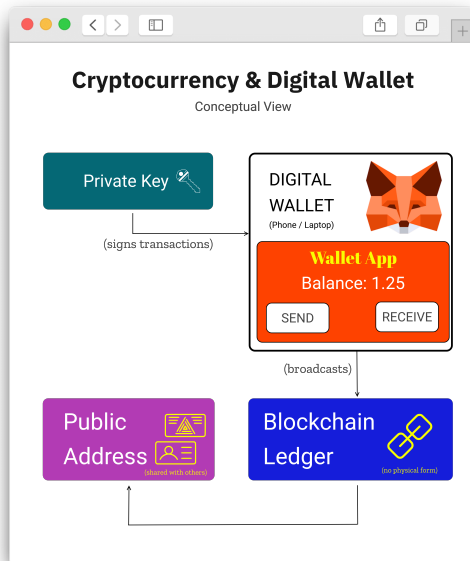
Cryptocurrency Visualized

Key components:

- ▶  Digital wallet (like bank app)
- ▶  Private key (your password)
- ▶  Public address (your account number)
- ▶  Transactions (sending/receiving)

No physical form:

- ▶ Exists only as entries on blockchain
- ▶ Controlled by cryptographic keys



The Evolution of Money

1.  **Barter:** Trade goods directly (ancient times)
2.  **Commodity money:** Gold, silver, shells
3.  **Paper money:** Government-issued currency
4.  **Digital money:** Credit cards, PayPal (1990s+)
5.  **Cryptocurrency:** Decentralized digital money (2009+)

Each step: More convenient, less physical

Money Evolution Timeline

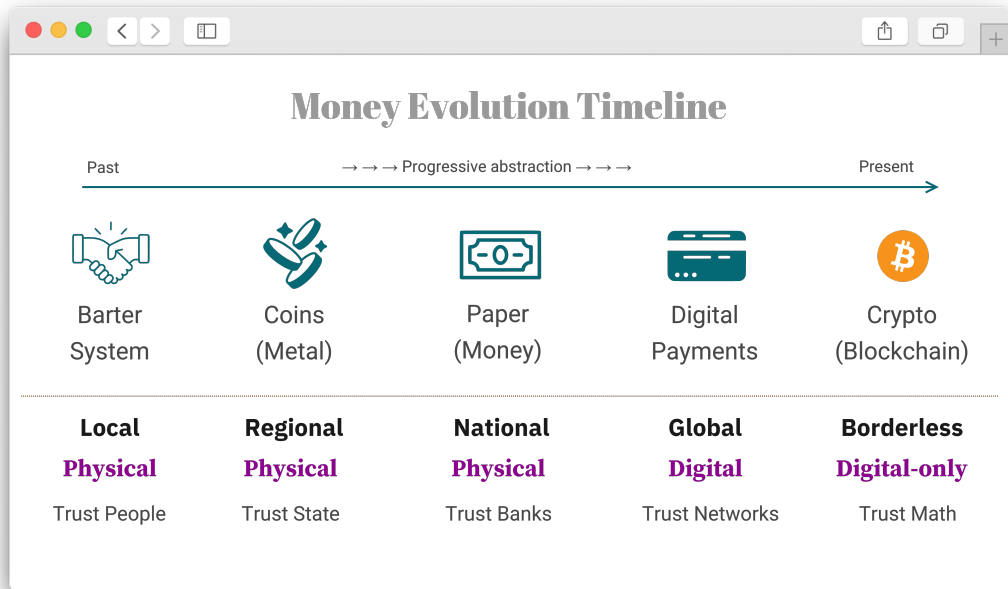
What changed:

- ▶ Physical → Digital
- ▶ Local → Global
- ▶ Centralized → Decentralized
- ▶ Trust institutions → Trust math

Key insight:

- ▶ Money is whatever people agree has value
- ▶ Crypto is the latest evolution

Why Was Cryptocurrency Invented?



Why Was Cryptocurrency Invented?

Problems with traditional money:

1. 🏦 Banks control your money (can freeze accounts)
2. 🌐 International transfers slow and expensive
3. 🖨️ Governments can print unlimited money (inflation)
4. 🕵️ Financial privacy concerns
5. 🚫 Billions of people lack access to banking

Bitcoin's solution: Money without middlemen

Key Features of Cryptocurrency

1.  **Decentralized:** No single authority controls it
2.  **Pseudonymous:** Addresses not tied to real identity
3.  **Secure:** Protected by cryptography
4.  **Borderless:** Works anywhere with internet
5.  **Limited supply:** Many cryptos have fixed maximum (like gold)

Unlike traditional money:

- ▶ Can't be confiscated easily
- ▶ No one can print more (for most cryptos)

How Cryptocurrency Gains Value

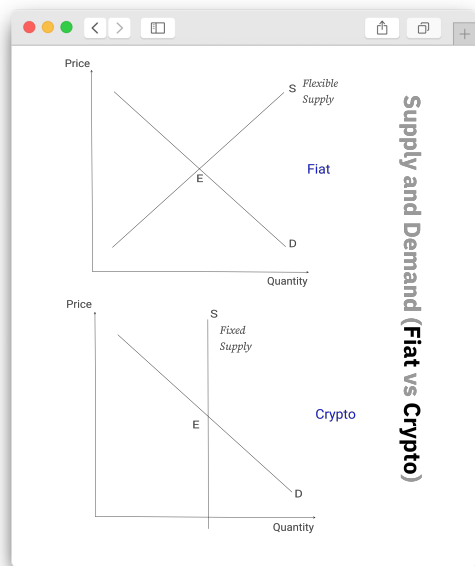
Basic economics:

- ▶  **Supply:** Limited number of coins
- ▶  **Demand:** People want to buy/use it
- ▶  **Utility:** What can you do with it?
- ▶  **Network effect:** More users = more valuable
- ▶  **Perception:** News and sentiment matter

Example:

- ▶ Bitcoin = 21 million coins maximum (fixed supply)
- ▶ More people want it → Price goes up

Supply and Demand Visualization



Fixed supply cryptos:

- ▶ Bitcoin: 21M max
- ▶ Supply can't increase
- ▶ Demand increases → Price up

Traditional money:

- ▶ Central bank prints more
- ▶ Supply increases
- ▶ Value can decrease (inflation)

Benefits of Cryptocurrency

1.  **Global access:** Anyone with internet can use
2.  **Lower fees:** Especially for international transfers
3.  **Security:** You control your funds (not a bank)
4.  **Fast:** Transactions settle in minutes, not days
5.  **Investment:** Potential for appreciation

Real-world example:

- ▶ Send \$1000 to Philippines: Bank = \$50 fee + 3 days
- ▶ With crypto: \$5 fee + 10 minutes

Drawbacks of Cryptocurrency

1.  **Volatility:** Prices swing wildly (risky)
2.  **Complexity:** Hard for average person to use
3.  **Irreversible:** Wrong address = money lost forever
4.  **Crime:** Used for illegal activities (ransomware)
5.  **Energy:** Bitcoin mining uses massive electricity

Important:

- ▶ If you lose your private key, you lose your money
- ▶ No customer service to call!

Traditional Money vs Cryptocurrency

Feature	Traditional (Fiat)	Cryptocurrency
Control	Central bank/government	Network consensus
Privacy	Bank knows everything	Pseudonymous
Borders	Restricted transfers	Borderless
Reversibility	Can reverse fraud	Irreversible
Inflation	Government can print	Usually fixed supply
Acceptance	Universal	Growing

Types of Cryptocurrency

1.  **Currency coins:** Bitcoin - digital gold, store of value
2.  **Platform coins:** Ethereum - build apps on blockchain
3.  **Stablecoins:** USDC, USDT - pegged to \$1 USD
4.  **DeFi tokens:** Used in decentralized finance apps
5.  **Utility tokens:** Power specific blockchain services

Next class: Deep dive into major cryptocurrencies

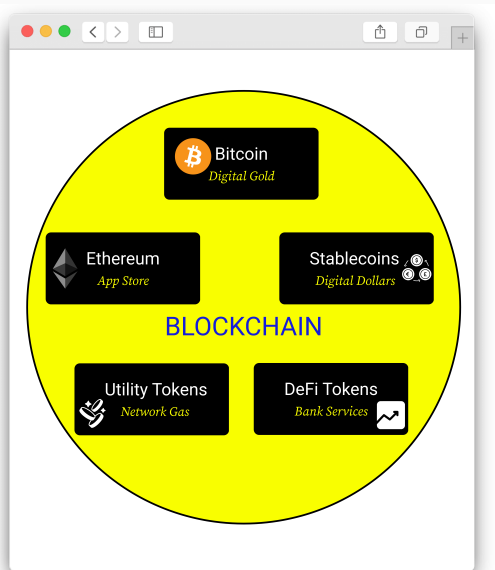
Crypto Categories Visual

Think of it like:

- ▶  Bitcoin = Digital gold
- ▶  Ethereum = App store
- ▶  Stablecoins = Digital dollars
- ▶  DeFi tokens = Bank services
- ▶  Utility tokens = Gas for network

All use blockchain:

- ▶ But serve different purposes



How Do You Get Cryptocurrency?

1.  **Buy on exchange:** Coinbase, Binance, Kraken
2.  **Receive as payment:** For goods or services
3.  **Earn:** Work, airdrops, rewards
4.  **Mining:** Validate transactions (advanced/expensive)
5.  **Swap:** Trade one crypto for another

Most common: Buy on exchange with credit card or bank transfer

Storing Cryptocurrency: Wallets

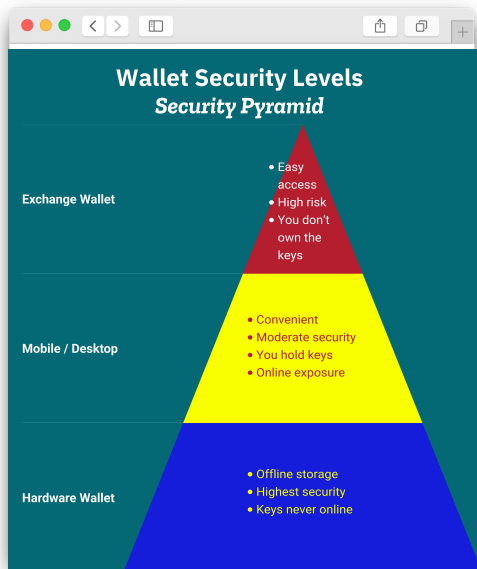
Types of wallets:

1.  **Exchange wallet:** Kept on Coinbase/Binance (easy)
2.  **Mobile wallet:** App on your phone (MetaMask)
3.  **Desktop wallet:** Software on computer
4.  **Hardware wallet:** Physical device (most secure)
5.  **Paper wallet:** Keys printed on paper

Golden rule:

- ▶ “Not your keys, not your crypto”
- ▶ If exchange holds it, you don't fully control it

Wallet Security Levels



Trade-offs:

- ▶  Exchange: Easy but risky
- ▶  Mobile: Convenient, moderate risk
- ▶  Hardware: Secure but less convenient

Recommendation:

- ▶ Small amounts: Mobile wallet
- ▶ Large amounts: Hardware wallet
- ▶ Never keep life savings on exchange

Understanding Crypto Addresses

What is a crypto address?

- ▶  Like your bank account number, but public
- ▶ Long string of letters and numbers
- ▶ Example: 0x742d35Cc6634C0532925a3b8...
- ▶ Anyone can send to it
- ▶ Only you can spend from it (with private key)

Important:

- ▶  Public address = Safe to share
- ▶  Private key = NEVER share (like password)

Private Key = Your Money

Critical concept:

- ▶  Private key = Proof you own the crypto
- ▶ Whoever has the key controls the funds
- ▶ Lost key = Lost money (permanently!)
- ▶ No “forgot password” option

Best practices:

1. Write down backup seed phrase (12-24 words)
2. Store in secure location (fireproof safe)
3. Never store digitally (no screenshots!)
4. Never share with anyone

Hands-On Activity: Compare Payment Systems

Let's compare sending money:

1.  Open two browser tabs
2.  Tab 1: Google “Western Union fees to send \$500”
3.  Tab 2: Check current Bitcoin transaction fee on Blockchain.com
4.  Compare: Fee amount + time to send
5.  Note: Which is cheaper? Which is faster?

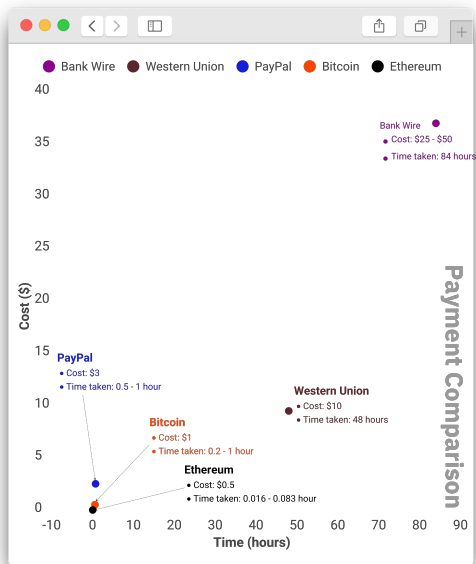
We'll discuss findings as a class

Payment Comparison Results

Typical findings:

- ▶ 🏦 **Bank wire:** \$25-50, 2-5 days
- ▶ 🏧 **Western Union:** 3-8%, 1-3 days
- ▶ 💳 **PayPal:** 2.9% + \$0.30, instant*
- ▶ ₿ **Bitcoin:** \$1-5, 10-60 min
- ▶ ⬡ **Ethereum:** \$0.50-3, 1-5 min

*Within same country



Transaction Fees Explained

Why do crypto fees vary?

- ▶  Network congestion (more users = higher fees)
- ▶  Transaction priority (pay more = faster)
- ▶  Which blockchain you use (Ethereum vs Solana)
- ▶  Time of day (weekends often cheaper)

Who gets the fees?

- ▶ Miners/validators who process transactions
- ▶ Incentive to secure the network
- ▶ Unlike banks, distributed to many participants

Real-World Crypto Use Cases

1.  **Remittances:** Sending money home to family abroad
2.  **Merchant payments:** Some stores accept Bitcoin
3.  **Investment:** Hold as store of value
4.  **Donations:** Support causes globally
5.  **Financial freedom:** Countries with unstable currency

Growing adoption:

- ▶ El Salvador made Bitcoin legal tender (2021)
- ▶ Major companies accept crypto (Microsoft, Tesla, AT&T)

Crypto Myths vs Reality

- ▶ **✗ Myth:** Only for criminals
 - ▶ **✓ Reality:** All transactions public and traceable
- ▶ **✗ Myth:** It's too late to get involved
 - ▶ **✓ Reality:** Still early adoption phase
- ▶ **✗ Myth:** You need to buy whole coins
 - ▶ **✓ Reality:** Can buy \$10 worth (fractional)
- ▶ **✗ Myth:** Completely anonymous
 - ▶ **✓ Reality:** Pseudonymous (can be traced)

Key Takeaways

1.  Cryptocurrency = Digital money on blockchain
2.  Value comes from supply, demand, and utility
3.  Benefits: Fast, global, lower fees vs Drawbacks: Volatile, complex
4.  Private key = Your money (guard it carefully!)
5.  Use cases: Remittances, payments, store of value

Remember:

- ▶ Never invest more than you can afford to lose
- ▶ Do your own research (DYOR)

Questions and Discussion